



Mini AI Computer T906 Datasheet

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Introduction

Before you use this manual, please read the following license agreement carefully. You may only use the products described in this manual if you agree to the following licence agreement.

Escape Clause

This document is based on current information and its contents are subject to change without notice. We have made every effort to ensure that its contents are accurate and reliable at the time of writing. But we accepts no liability for loss or damage arising from omissions, inaccuracies, or errors in this document.

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For more information, you can also refer to [NVIDIA official Jetson Download Center](#)

Safety Warnings and Precautions for Use

● Security instructions

Before using this product, you must review this document to have a preliminary understanding of the product, and must follow the safety instructions in this product manual to ensure your personal safety and to avoid damage to the equipment. The manufacturer is not responsible for any problems caused by the wrong operation of the equipment and the safety of personal life and property.

● Supply voltage

Mini AI Computer T906 edge computing platform has a stable and reliable power supply of typical power 45W maximum power 75W at the input.

Power supply range: 9- 36 V

● Environmental requirements

Operating temperature: -20°C - 65°C

Ventilation requirements: The perimeter of the calculation platform installation must be well ventilated.

● Grounding requirements

The power supply of the power adapter must be well grounded, special cases need to install the grounding screw grounding on the calculation platform.

● Static Protection

Electronic components and circuits are sensitive to electrostatic discharge. Although we design our circuit board card products with anti-static protection for the main Interfaces on the board, it is difficult to achieve anti-static safety protection for all components and circuits. Therefore, when handling any circuit board components, it is recommended that anti-static safety measures be observed. Anti-static safety protection measures include, but are not limited to, the following.

- ◆ The box should be placed in an anti-static bag during transportation and storage until the board is installed and deployed, then take out the card.
- ◆ The static electricity stored in the body should be discharged before the body touches the box: wear a discharge grounding wristband.
- ◆ Operate the box only in the safe area of the static discharge point.
- ◆ Avoid moving boxes in carpeted areas.

● Operation and Maintenance

Operators or maintenance personnel must be trained and qualified before they can participate in operation or maintenance.

Brief Introduction

Mini AI Computer T906 is an edge AI supercomputer for unmanned field. Combines Jetson AGX Orin 32GB module, deliver up to 200 TOPS of AI performance, with Pre-installed JetPack 5.0.2, 10GbE and 1GbE, with a rich set of IOs— HDMI, USBs, CAN.GPIO, RS232, AUDIO, Support GPS, WiFi and 4G/5G.

It is designed with advanced heat dissipation, lightweight and pre-installed bottom bracket for easy field installation, long MTBF stable operation capability, it is an ideal device ready for developing advanced robots, autonomous machine products, unmanned delivery vehicles, unmanned cleaning vehicle, low-altitude defense, intelligent inspection, intelligent buildings and other autonomous machines.

Mini AI Computer T906 Edge Computing Platform Overview

- **Robotics-suited edge AI box:** Advanced heat dissipation, JAE Automotive Grade IO Plugs and rich Industrial IO interface types
- **Giant Leap forward AI module:** NVIDIA® JETSON AGX ORIN™ module, Delivers 200 TOPS AI performance, supports multiple concurrent AI inference pipelines, plus high-speed interface support for multiple sensors.
- **Industrial connectivity and Rich set of I/Os:** 3xRS232, 3xCAN, 2xGPIO, AUDIO, 4xUSB 3, 1xHDMI, M.2 KEY M
- **Hybrid network access:** 1x 10GbE, 1x GbE port, Support Wifi, Bluetooth and 5G/4G(modules not included).
- **Advanced heat dissipation:** Two fans combined with passive heat dissipation
- **Pre-installed NVIDIA JetPack 5.0.2:** support the entire Jetpack software stack and various developer for building fast and robust AI application provided by Sseed Edge AI partners, helps develop innovative AI solution for manufacturing, logistics, retail, service, agriculture, smart city, healthcare, and life sciences, etc



Product Specifications

Processors and Core Modules

Processor	NVIDIA AGX ORIN
CPU	8-core NVIDIA Arm® Cortex A78AE v8.2 64-bit CPU 2MB L2 + 4MB L3
CPU Max Frequency	2.2 GHz
GPU	1792-core NVIDIA Ampere GPU with 56 Tensor Cores
GPU Max Frequency	939 MHz
Memory	32 GB 256-bit LPDDR5
DL accelerator	2 x NVDLA v 2.0
DLA Max Frequency	1.4 GHz
Storage	64GB eMMC 5.1 1 x m.2 key m nvme 2280 (module not included)

CODEC (coder and decoder)

Video Encode	1x 4K60 (H.265) 3x 4K30 (H.265) 6x 1080p60 (H.265) 12x 1080p30 (H.265)
Video Decode	1x 8K30 (H.265) 2x 4K60 (H.265) 4x 4K30 (H.265) 9x 1080p60 (H.265) 18x 1080p30 (H.265)

Interface

	Interface	Quantity	Note
Network	Ethernet	2×RJ45 Gigabit Network port	1x 10GbE 1x 1GbE
	WIFI	1	Module not included
	4G	1	Module not included Only 4G or 5G at one time
	5G	1	
Video output	HDMI	1x HDMI 2.0	Does not support adapter signals (such as vga to hdmi form)
USB	USB	4×USB 3.0TYPE A	USB 5V, 1A
	USB-OTG	1xUSB2.0 Type-C	
	USB-debug	1xUSB Type-C	
I/O	GPIO	2xGPIO	Independent 3.3V TTL level GPIO
	CAN	3xCAN 2.0b	With CAN chip
	RS-232	2x RS-232 D-SUB9 1 xTIME SYNC(RS232)	DB9 Terminal
User Expansion	M.2	1 x M.2 M Key	PCIE NVME 2280 SSD Module not included
Function Key	Power KEY	1	Button
	Reset KEY	1	Button
	Recovery KEY	1	Button

Modules Connection

Modules	Interface
WiFi	B Key
SSD	M Key
4G	Mini PCIe
5G	E Key
GPS	Surface-mount

Supply Electricity

Power Supply	Spec
Input Type	JAE Connector MX23A12SF1
Input Voltage	Wide input 9-36V DC
Typical consumption	Typical power 45W maximum power 75W

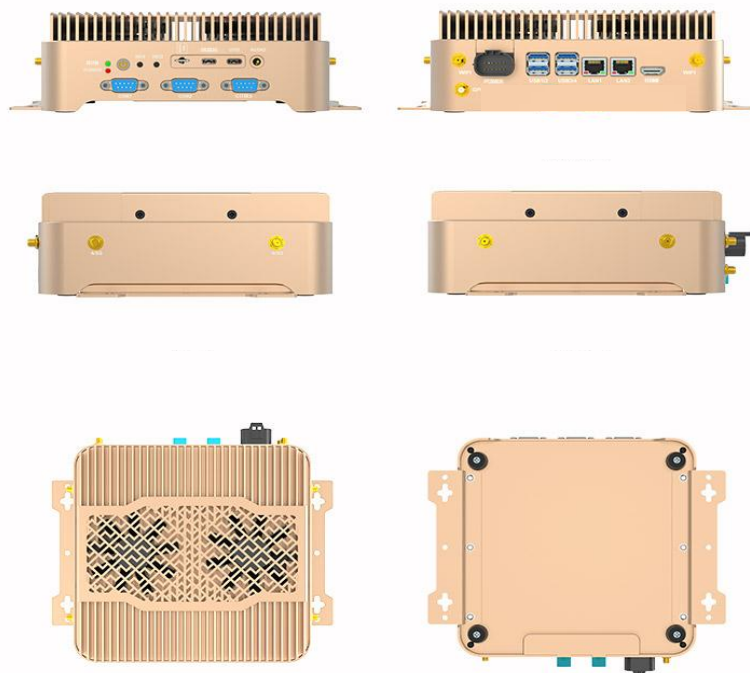
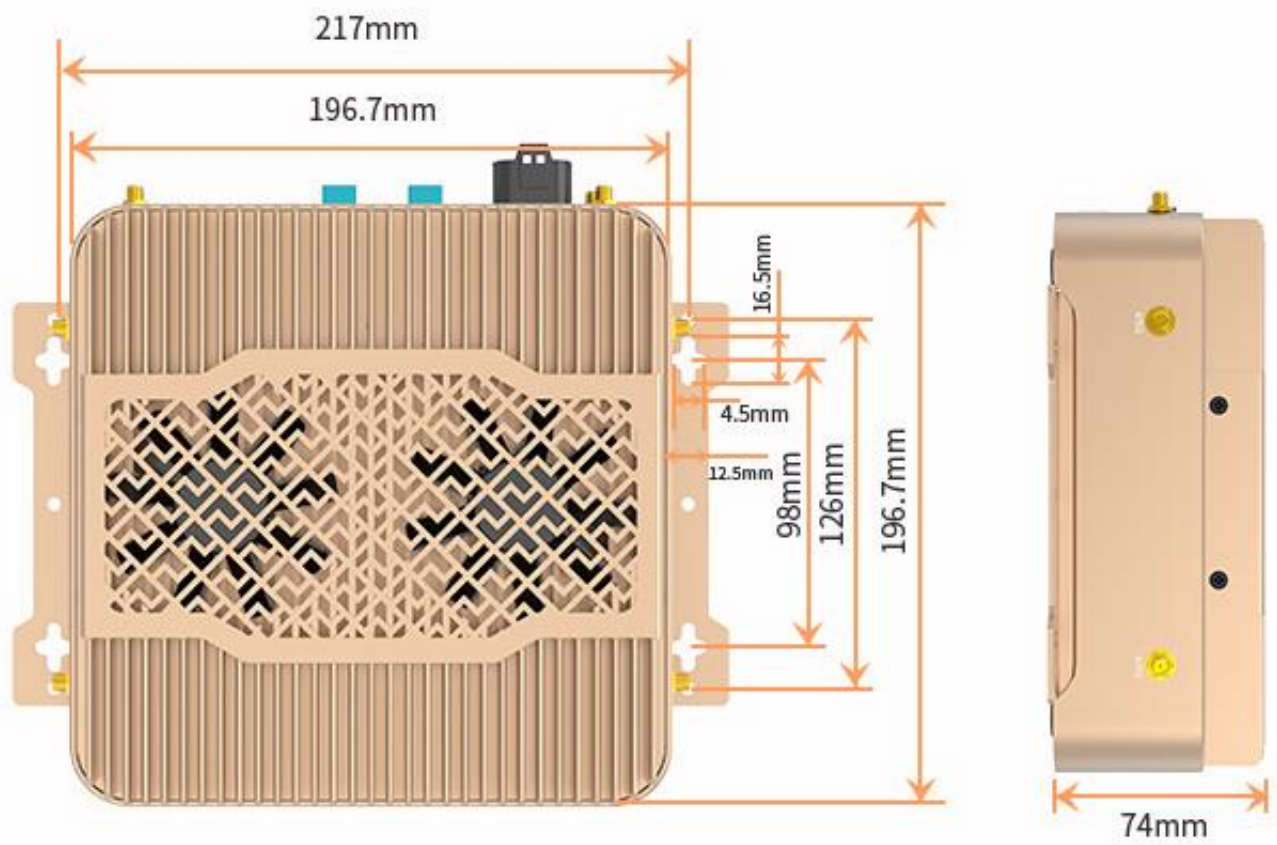
Structure

Mechanical	Spec
Dimensions (W×D×H)	196.7*196*74mm
Weight	2.5Kg

Environments

Environmental	Spec
Operating Temperature	-20°C to 65°C
Storage Humidity	10%-90% non-condensing
Protection	IP55

Size



Description of Interface

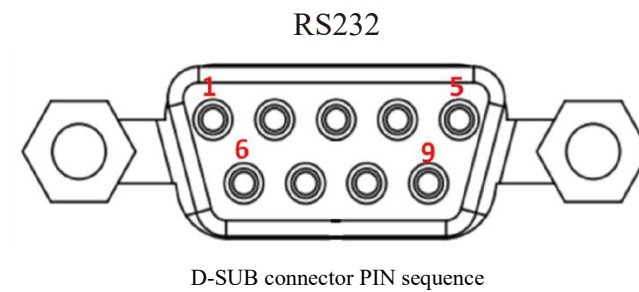
Front Interface



Interface	Name of Interface	Description of Interface
POWER	Power button	Press and hold to turn off the computer, press the screen to display four related options
RES	Restore button	Use alone for reset and reboot, with REC to enter recover mode
REC	Reset button	No effect alone, use with reset button to enter recover mode
SIM	SIM card slot	Install the SIM card in the direction shown
DEBUG	Debug	Output debug information by connecting TYPE-C cable
USB-OTG	TYPE-C Interface	Data can be exchanged by connecting a normal usb cable device
POWER_LED	Power Status Indicator	When the power is turned on: the indicator light is green and always on
RUN_LED	System status indicator	When the power is turned on: the indicator light is red and always on
AUDIO	Audio Interface	
COM1	RS232	Standard RS232 serial port (device number is ttyTHS0)
COM2	RS232	Standard RS232 serial port (device number ttyTHS1, if using GPS, this interface is occupied)
COM 3 TIME SYNC	RS232	Standard RS232 serial port (device number is ttyTHS4)

Note: Recover mode refers to the download mode, which is mainly used to reinstall the system and install part of the sdk using SDK manager. Press and hold the REC button, then press and hold the RES button. 2 seconds later, release the RES button, and finally release the REC button. Enter lsusb in the host terminal and check if there is an nvidia corp. if there is, the device is successfully entered, if not, check if the type-c cable is connected, if the host usb is connected, and if the sequence and duration of key presses are correct. For the host usb interface, we recommend that you use the usb3.0 interface.

Serial Port Description



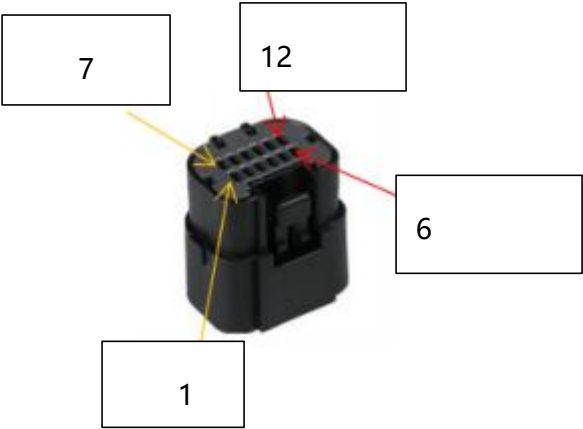
RS-232 Connector PIN Definition			
PIN#	Meaning	PIN#	Meaning
1	DCD	2	RXD
3	TXD	4	DTR
5	GND	6	DSR
7	RTS	8	CTS
9	RI		

Back Interface



Interface	Interface Name	Interface Description
I/O	Multifunctional interface	Includes CAN0 and CAN1, 2xGPIO and power supply inputs
WIFI	WIFI Antenna Interface	External WIFI antenna
4/5G	5G Antenna Interface	External 5G antenna
GPS	GPS Antenna Interface	External GPS antenna
HDMI	HDMI Interface	Standard HDMI 2.0Interface
LAN1	10GbE	
LAN2	1GbE	
USB	USB 3	4 xUSB 3

Multifunctional interface

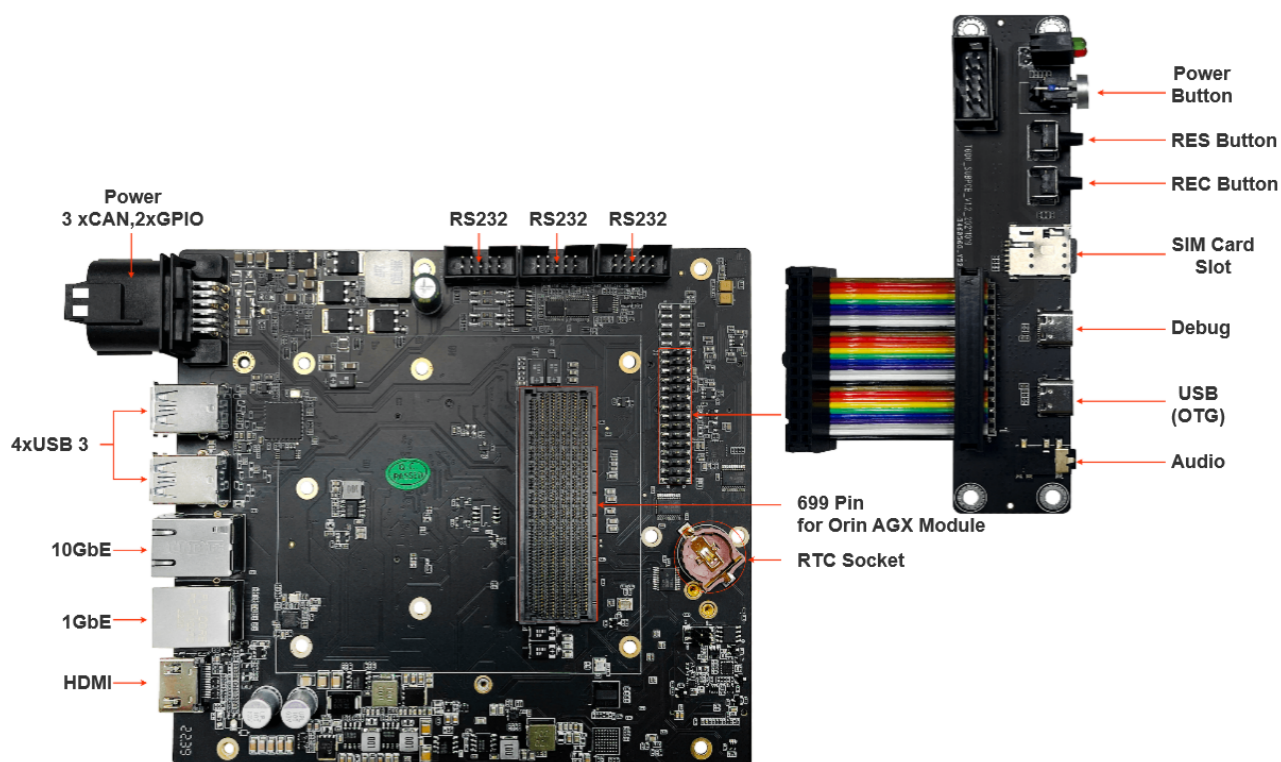


1	2	3	4	5	6
SPI-CAN-H0	CAN-H0	CAN-H1	GND	GPIO1	VCC-IN
7	8	9	10	11	12
SPI-CAN-L0	CAN-L0	CAN-L1	GND	GPIO2	VCC-IN

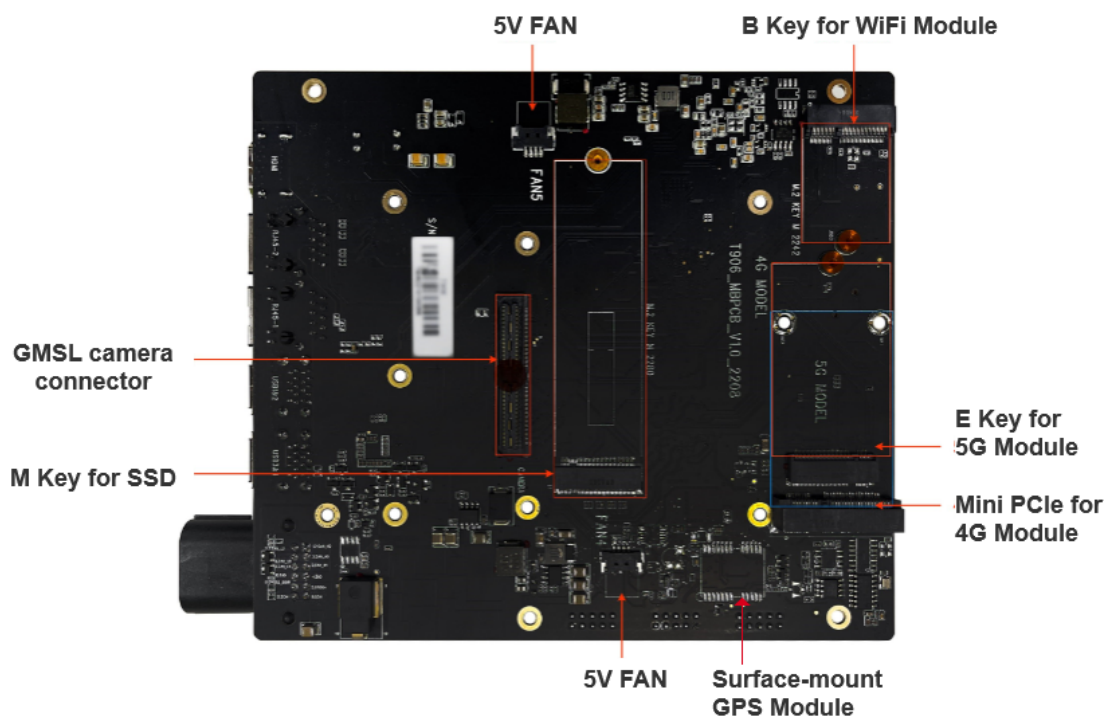
Serial Port Description

PIN			
PIN#	Meaning	PIN#	Meaning
1	SPI-CAN-H0	7	SPI-CAN-L0
2	CAN_H0	8	CAN-L0
3	CAN-H1	9	CAN_L1
4	GND	10	GND
5	GPIO1 (GPIO-350 PA.02 0-1.8V) Default is output, low	11	GPIO2 (GPIO-390 PG.07 0-3.3V) Default is output, low
6	VDDIN(12-24V)	12	VDDIN(12-24V)

Carrier Board interface



RTC Connector for Rechargeable Battery
Battery included



4G, 5G, WiFi, GPS modules not included

System

System Description

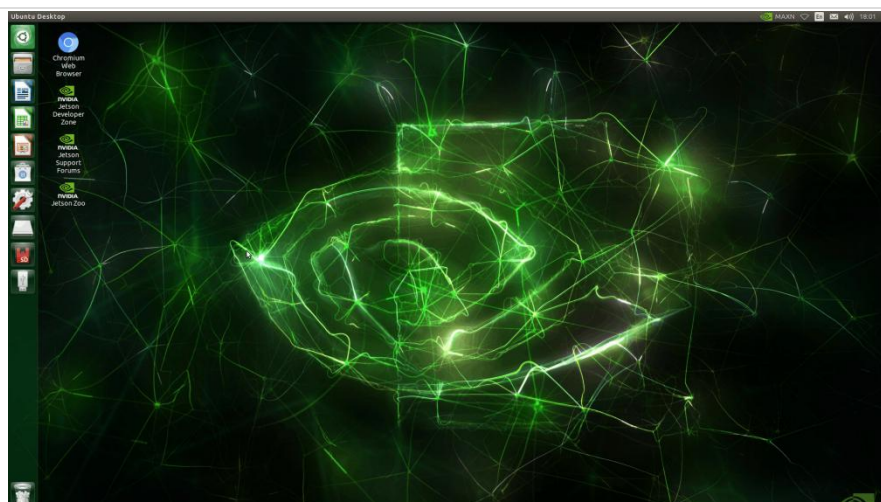
Mini AI Computer T906 device is powered by Jetpack 5.0.2. Default username: nvidia Password: nvidia

System Rebrush

Please refer to the driver installation section, which describes in detail how to brush and how to do it.

Startup & Shutdown

Power On: The default power-on mode of the Mini AI Computer T906 unit is power-on self-start. Plug in the power and connect the monitor to the unit via HDMI Interface, the power-on screen is as shown in the figure.



Boot screen

Shutdown: Press and hold the POWER key to shutdown, or execute `$ sudo poweroff` at the command line.

Reboot: execute `sudo reboot` in the command line to complete the reboot; you can also reboot by pressing the reset key (just press it)

Part list



Mini AI Computer T906 x1



**19V power adapter x1
(without power cord)**



Mounting bracket x1



Type-C cable x1



Power adapter cable x1